```
from collections import deque
q = deque([((3, 3, 0), [])])
moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]
seen = set()
while q:
    (m, c, b), path = q.popleft()
    if (m, c, b) in seen:
        continue
    seen.add((m, c, b))
    if (m, c, b) == (0, 0, 1):
        print(path + [(m, c, b)])
        break
    for x, y in moves:
        nm = m - x if b == 0 else m + x
        nc = c - y if b == 0 else c + y
        if 0 <= nm <= 3 and 0 <= nc <= 3:
            if (nm == 0 or nm >= nc) and (3 - nm == 0 or 3 - nm >= 3 - nc):
                q.append(((nm, nc, 1 - b), path + [(m, c, b)]))
```

OUTPUT:

```
[(3, 3, 0), (3, 1, 1), (3, 2, 0), (3, 0, 1), (3, 1, 0), (1, 1, 1), (2, 2, 0
), (0, 2, 1), (0, 3, 0), (0, 1, 1), (1, 1, 0), (0, 0, 1)]
```